.

Forty Years of Overset CFD



Stuart Rogers

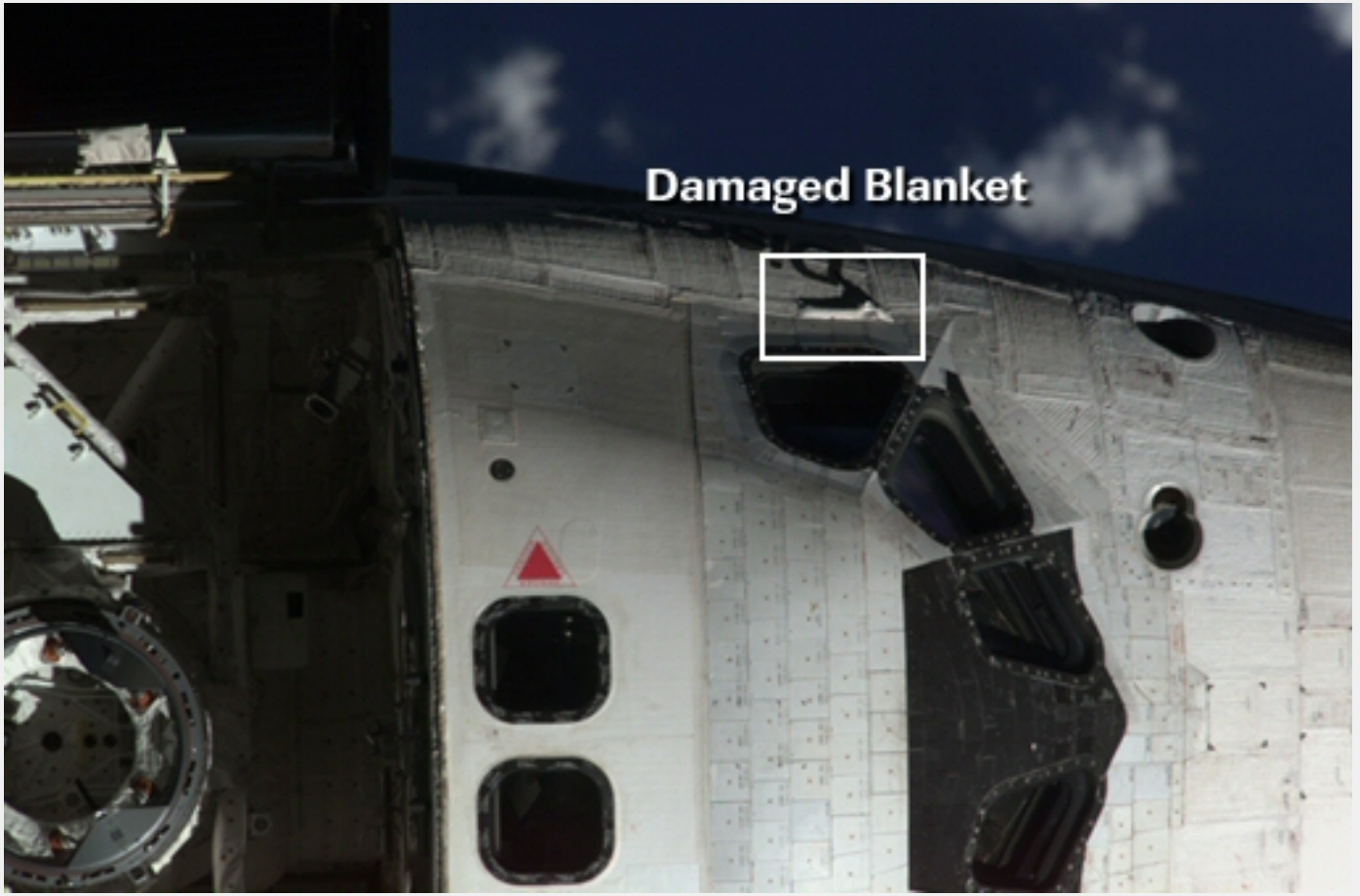
2026 Overset Grid Symposium

Penn State

September 24, 2026

Prologue

August 3rd, 2005: A Pretty Good Day at Work





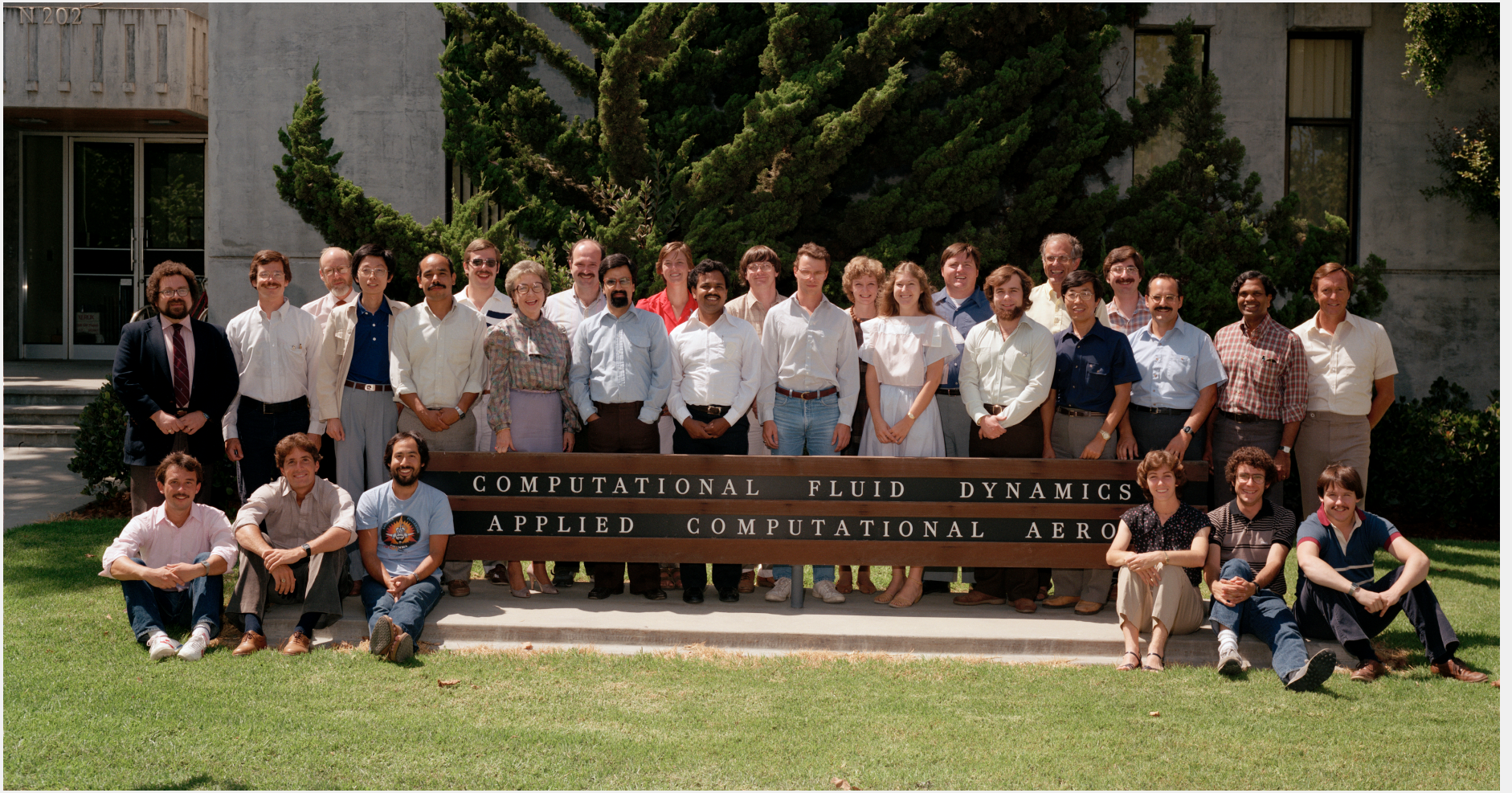
Introduction

For the past 34 years, the Overset Grid Symposium has been an outstanding venue leading to interactions between different developers and users that has helped drive ideas, software development, and applications.

- *Prologue*
- **NAS: The Early Years**
  - Building 202A: 1983-1985
  - NAS Design
  - SGI Workstations
- Advanced Subsonics Technology Program: 1990s
- Space Shuttle Program: 2002-2011
- Orion Aerodynamics: 2012-2014
- Space Launch System: 2014-2026
- Lessons and Observations

Building 202A: 1983 - 1985

- Intern M.S. Program University of Colorado
- Incompressible Navier-Stokes group
- Funded by MSFC Space Shuttle Main Engine project
- NAS: Numerical Aerodynamic Simulator
- Building 258: moved in early 1986



Growth of CFD at Ames

Ames line managers in the late 1980s included many CFDers

- Branch Chiefs: Harv Lomax, Paul Kutler and then Terry Holst, Tom Pulliam
- Division chief Paul Kutler
- Directorate chief Ron Bailey
- Ames center director Bill Ballhaus
- NAS Cray 2 Supercomputers were named Navier and Stokes

JSC/Ames Overset CFD Collaboration

Space Shuttle funded effort for technology development

- Starting in 1987

- Ames group led by Joe Steger
    - Bob Meakin, Pieter Buning, William Chan, ...

- JSC group led by Fred Martin
    - Jeff Slotnick, Ray Gomez, Jay Le Beau, ...

- See Buning and Gomez Invited talk OGS 2010



Shuttle Group at KSC in 1988: Riba, Meakin, Chiu, Buning, Rizk, Obayashi, Steger, Slotnick, Martin

NAS Design

NASA TM 84-386:

- V.L. Peterson
- W.F. Ballhaus
- F.R. Bailey

Subsystems:

- High-Speed Processor
- Support Processing
- Workstation
- Graphics
- Mass Storage
- Long-Haul Communications
- High-Speed Data-Network

Environment:

- Common user environment
- Unix operating systems

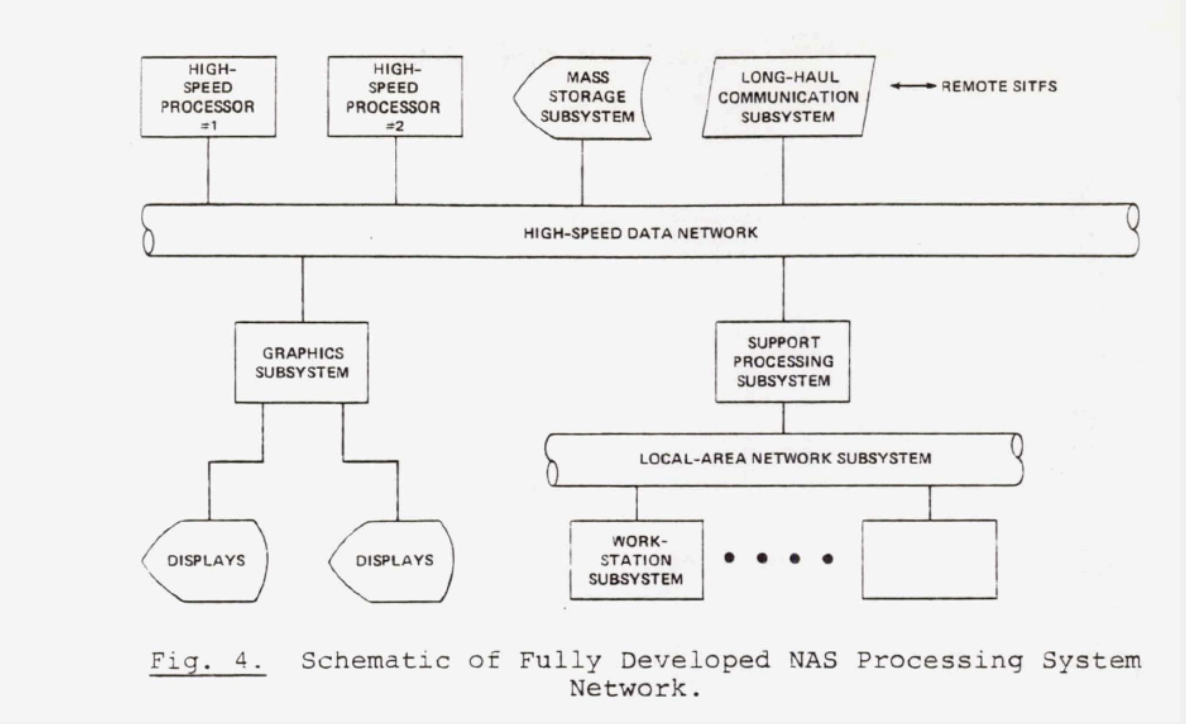


Fig. 4. Schematic of Fully Developed NAS Processing System Network.

NAS Schematic

Cray 2



SGI Workstations

Early NAS SGI workstations in Building 202A in 1984-1985

- Graphics only available on VAX/VMS VT100 terminals
- IRIS 1200 terminals
- IRIS 2400 first workstations in 1984/85 in 202A: rgb cables
- Pieter Buning: Porting of plot3d to run on SGI native graphics
- GL2 OS -> IRIX SGI unix-like operating system
- Upgrades to 2500, 2500T (turbo), and 3000
- IRIS 4D/70, 4D/70 GT
- MEX (multi-exposure) windowing system





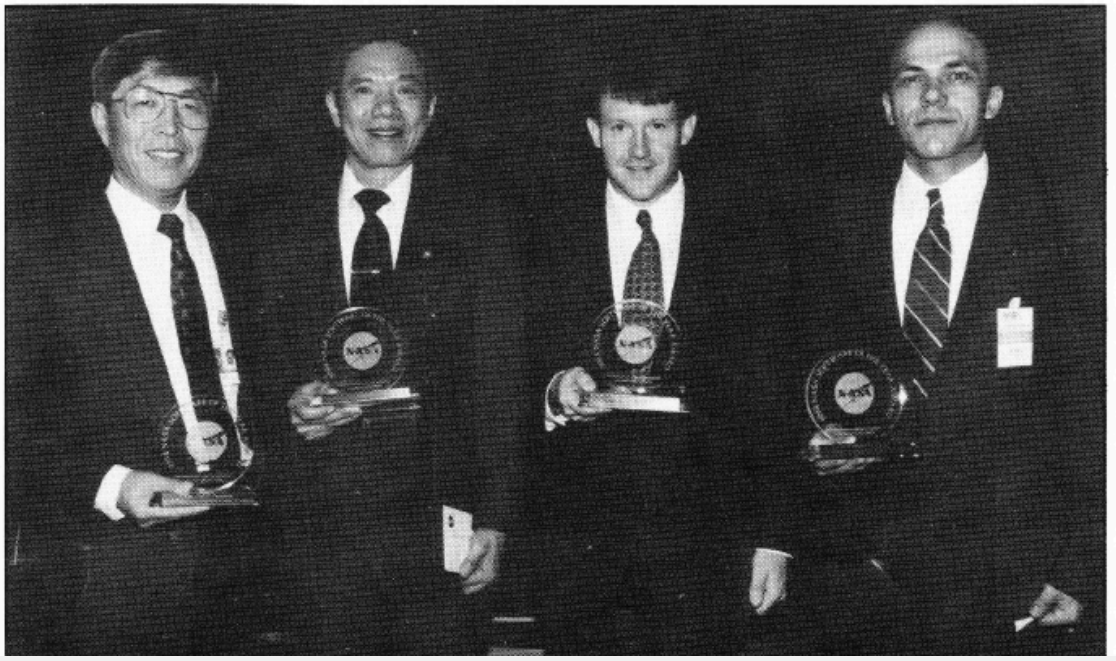
1994 Astrogram

National Aeronautics and Space Administration

Astrogram

Ames Research Center, Moffett Field, California — January 13, 1994

Researchers at Ames Win New Prestigious NASA Software Award



Ames Researcher Chawla Named Astronaut Candidate

By Pamela Davoren

"Are you still interested in coming down and working for us?"

With those words, Ames researcher Dr. Kalpana Chawla was awakened early Dec. 7 by a telephone call from former mission specialist David Leestma in Houston. She had been accepted as an astronaut candidate.

That day truly 'will live in memory' for Chawla. Although Chawla advocates "dare to dream," she needed the NASA press release and congratulatory letter as written proof that she wasn't dreaming this time.

She will be joining the ranks of former Ames researchers Drs. Ellen Ochoa

1994 Astrogram

Kalpana Chawla Joins Fellow Ames Employees in the Astronaut Corps

(Continued from Page 1)

microgravity," explained Chawla.

Co-workers know Chawla has the "right stuff" and are very happy for her — "very qualified researcher, diligent, first-rate, very capable of doing a fine job."

Chawla's journey began 33 years ago in Karnal, India, where she was the youngest in a family with two sisters and one brother. "Kalpana" means imagination. Little did they know just how appropriate that name would be.

She credits her father ("a very technical-minded person…easily one of the smartest engineers I know") with being an influence. At

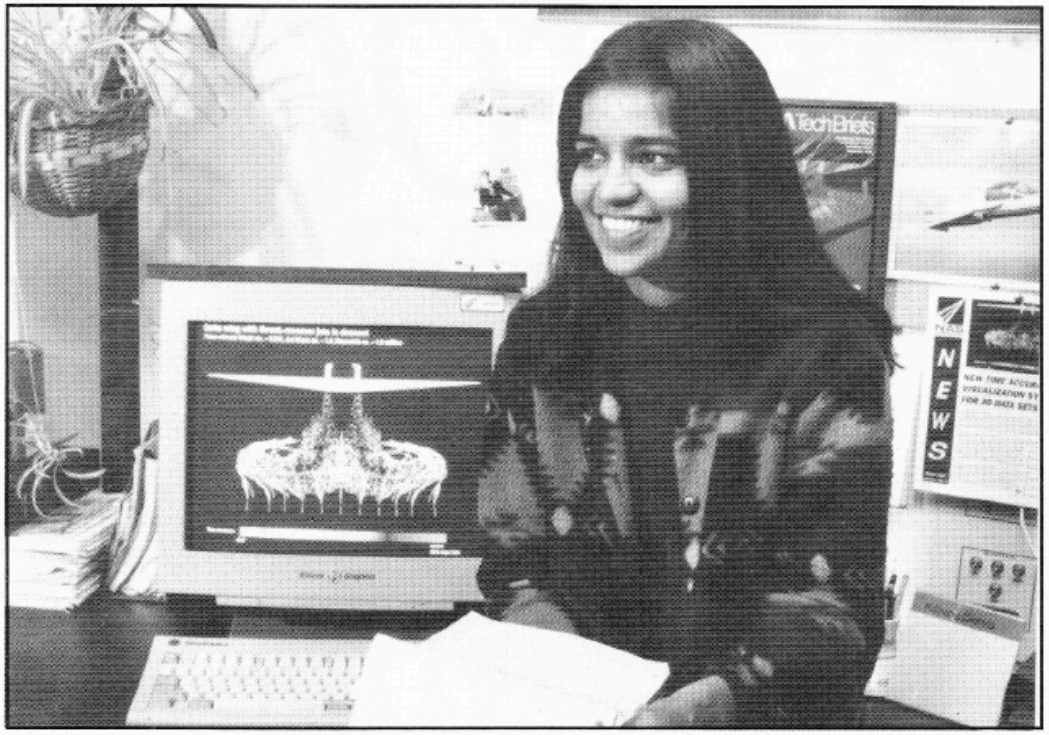


Photo — Cesar Acosta

Dr. Kalpana Chawla, seen in her office in Bldg. N-253, continues the tradition of Ames employees who have joined the ranks of the NASA astronaut corps. Soon she will be trading in her work with computational fluid dynamics for the dynamics of the comprehensive year-long training program of an astronaut candidate. Chawla is one of only six civilian mission specialists who were selected for the astronaut candidate class of '95. "I think I've been very fortunate," said Chawla.

capability to handle other technical problems," said Chawla.

Her solid academic and technical background, her love of aviation (she has nearly 650 flying hours and is working on her instructor's rating) and earning her U.S. citizenship, equipped Chawla with the confidence to apply as an astronaut candidate.

She will be bringing a wealth of talent, experience and background to Houston. Her flexibility and willingness to do a number of things will serve her well.

The personal traits that prepared her for this challenge? "I've always been very determined. I don't get

- *Prologue*
- *NAS: The Early Years*
- **Advanced Subsonics Technology Program: 1990s**
  - Collaboration between NASA and Industry
  - Integrated Wing Design: design-cycle tools
- Space Shuttle Program: 2002-2011
- Orion Aerodynamics: 2012-2014
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Advanced Subsonics Technology Program

- Moved offices to Unitary Wind Tunnel Building 227B

- Collaboration between LaRC, ARC, Boeing, McDonnell-Douglas

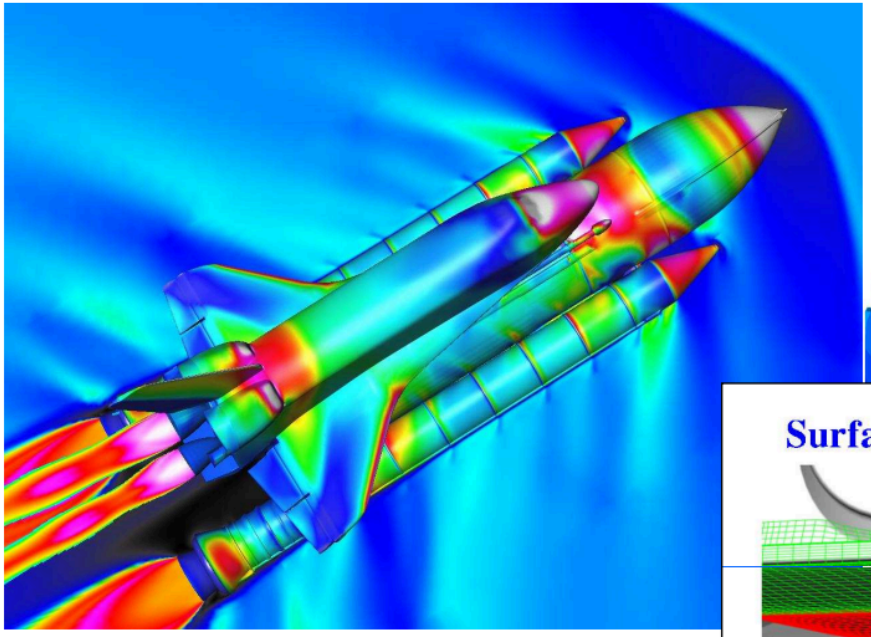
- Wind-tunnel testing
  - Ames 12-foot Pressure Wind Tunnel

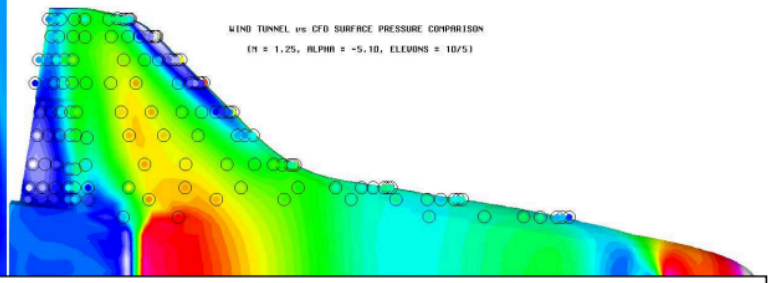
- CFD tool development
  - Turbulence model development
  - Chimera Grid Tools
  - Overflow
  - Pegasus5
  - Long-term impact on overset CFD

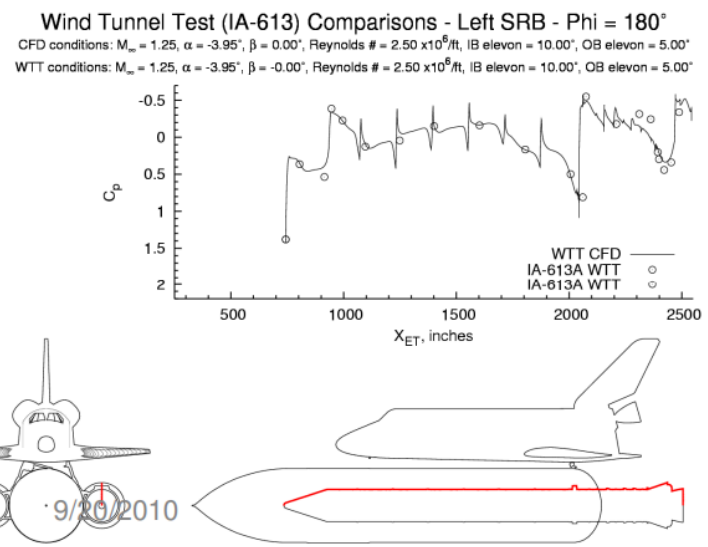
- High-Lift Aerodynamics
  - Multi-element airfoils
  - Trapezoidal Wing
  - High-Wing Transport: turbine-powered simulators
  - Boeing 777

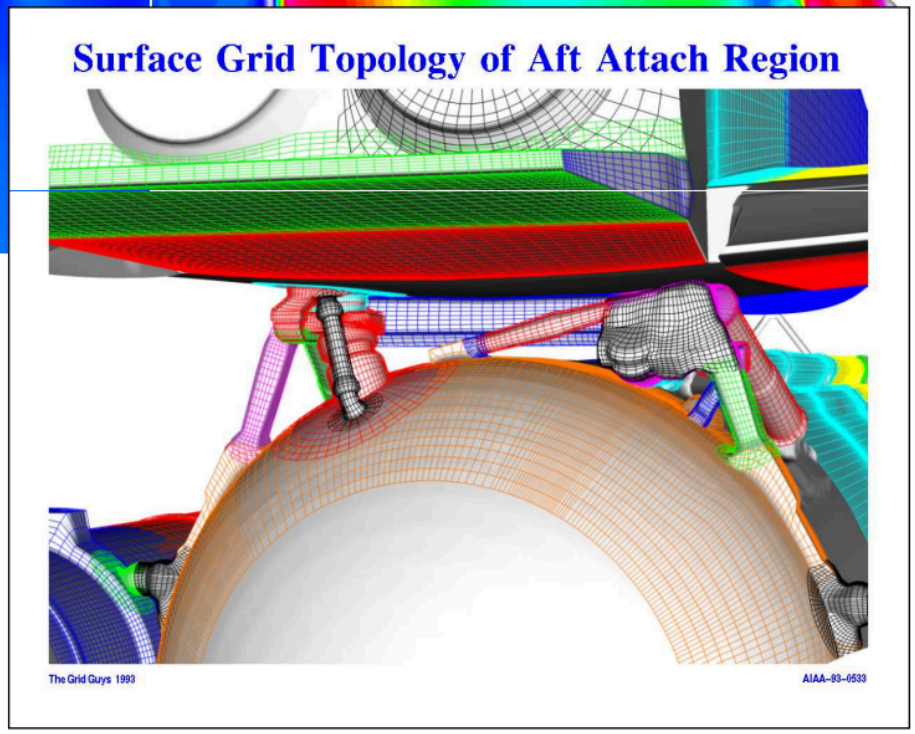
- *Prologue*
- *NAS: The Early Years*
- *Advanced Subsonics Technology Program: 1990s*
- **Space Shuttle Program: 2002-2011**
  - Columbia Accident: Foam
  - Return To Flight: Debris Transport
  - STS-114
  - Day-of-Launch Support
- Orion Aerodynamics: 2012-2014
- Space Launch System: 2014-2026
- Lessons and Observations

Overflow CFD of Space Shuttle Launch Vehicle



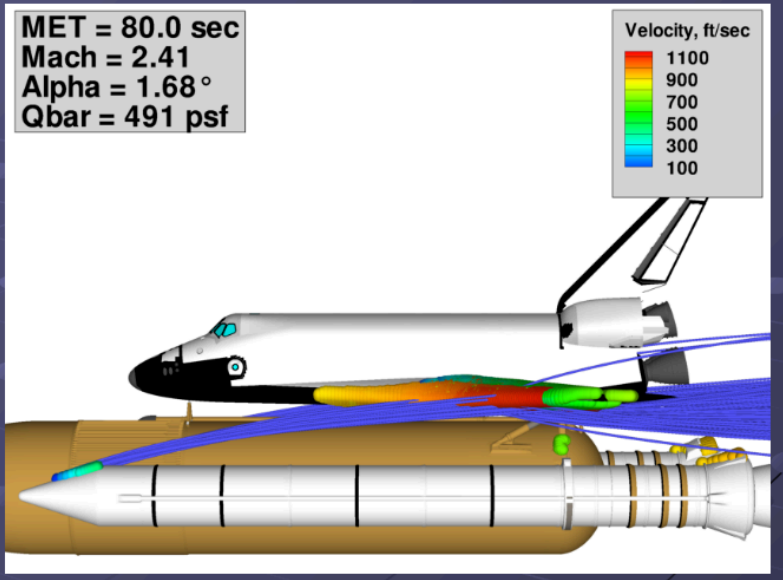


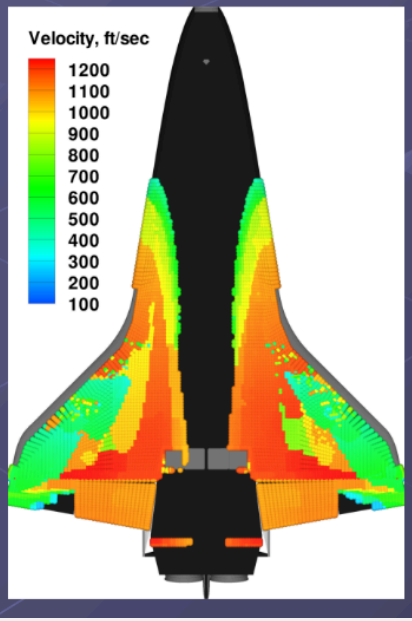


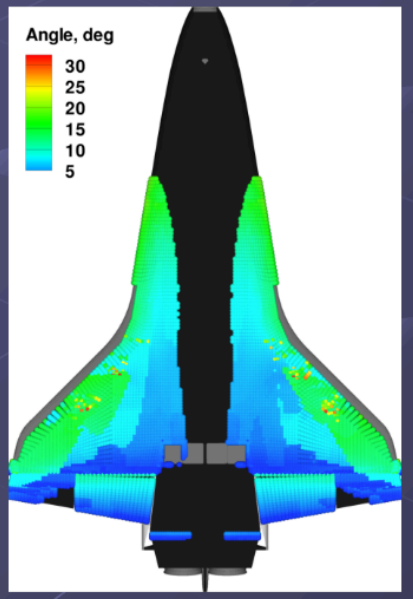


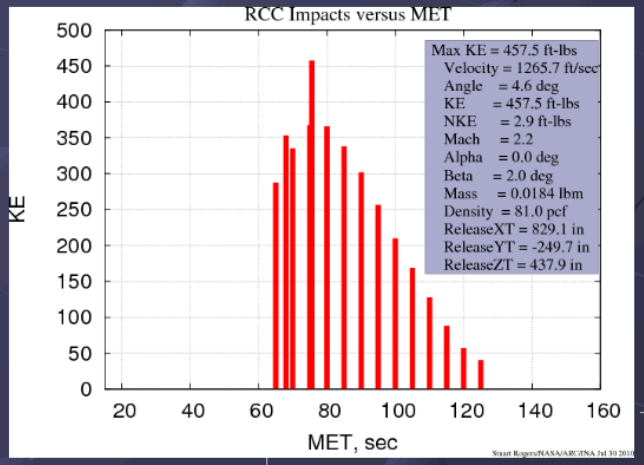
Debris Transport Using Overflow CFD











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STS-114 Mission: Return To Flight

- Tue July 26 10:39 EDT: launch
- Wed July 27 evening: MMT tasked debris team: analyze IFR foam debris
  - Ran DTA on NAS Columbia and reported by midnight PDT
  - no damage
  - consistent with vehicle inspections
- Thu July 28 morning: DTA results showed to Wayne Hale and MMT
- Wed Aug 3rd:
  - EVA #3: removal of 2 gap fillers
  - Ames WT test to analyze Orbiter torn blanket
- Thu Aug 4th: no repairs needed on torn blanket
  - No 4th EVA would be required
  - Astronaut Steve Robinson later recalled they were greatly relieved

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- *Prologue*
- *NAS: The Early Years*
- *Advanced Subsonics Technology Program: 1990s*
- *Space Shuttle Program: 2002-2011*
- **Orion Aerodynamics: 2012-2014**
  - Launch Abort Vehicle: plumes
  - Command Module: RCS jets
- Space Launch System: 2014-2026
- Lessons and Observations

- *Prologue*
- *NAS: The Early Years*
- *Advanced Subsonics Technology Program: 1990s*
- *Space Shuttle Program: 2002-2011*
- *Orion Aerodynamics: 2012-2014*
- **Space Launch System: 2014-2026**
  - Ascent Aerodynamics
  - Debris Transport
  - Day-of-Launch
  - Artemis I (Nov 2022) and Artemis II (April 2026)
- Lessons and Observations

- *Prologue*
- *NAS: The Early Years*
- *Advanced Subsonics Technology Program: 1990s*
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- *Orion Aerodynamics: 2012-2014*
- *Space Launch System: 2014-2026*
- **Lessons and Observations**

Lessons and Observations

- Bob MacCormack: be a detective
- Bill Ballhaus Jr.: hard work rewarded over time
- Errors are how you learn
- Own your mistakes
- Never too old to learn new stuff

Lessons and Observations

- Grit: the passion and perseverance required to achieve long-term, meaningful goals
  - Sustaining effort despite setbacks
  - Resilience, endurance, commitment, **HOPE**
- There are good lessons in the CAIB report
  - Understand *normalization of deviance*

Lessons from SLS Program Support

The Ames SLS Aero team has supported SLS for 12 years:

- There is no such thing as best practice, always room for improvement
- Delivering data to a flight program must buy their way on: prove that you can provide better data cost effectively
- Motivation should be based on helping the mission, not on parochial or personal gain
- Communication skills are as important as good data
- Hope you never have to hear this phrase: “But we have always done it that way”
- Know your customers
- Repeatability

Epilogue

To quote Nick Offerman as Ronald Ulysses Swanson:

> If you show up on time, speak honestly, and treat everyone with fairness, we will get along just fine